

Name: _____



Exploring the shapes of graphs using technology

Task A: Reviewing plotting graphs of equations

1) Create a table of values to plot the linear graph of $y = 11 - 5x$ for x -values in the range $-2 \leq x \leq 3$ in steps of 0.5

2)

a) Without using a table plot the linear graph of $y = 7 - x$

b) Without using a table plot the linear graph of $x + y = 5$

$$x + y - x = 5 - x$$

$$y = 5 - x$$

c) What do you notice about the shape of the two graphs?
Write a sentence.



Task B: Plotting quadratic graphs

1) Categorise these equations into **quadratic**, linear, and 'other'.

a) $y = x - 5$

b) $y = x^2 - 5$

c) $y = x^4 - 5$

d) $y = x^2 - x + 5$

e) $y = x^2 - 5x$

f) $y = x^2 - 5x + 1$

g) $y = 5 + 2x$

h) $y = 5 + x^2$

i) $y = x^5 + x^2$

j) $y = 5(x + 2)$

k) $y = x(x + 2)$

l) $y = x(x^2 + 2)$

Hint: Use Desmos to check if you're unsure. If the graph forms a **parabola** it is **quadratic**.

2) Use technology to plot the graphs of these quadratic **equations**. They all form **parabolas**. Spot the odd one out.

a) $y = x^2 + 3$

b) $y = 3 + x^2$

c) $y = x^2 - 3$

d) $y = 3 - x^2$